

EEGMatch: Learning With Incomplete Labels for Semisupervised EEG-Based Cross-Subject Emotion Recognition

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Abstract—Electroencephalography (EEG) is an objective tool for emotion recognition and shows promising performance. However, the label scarcity problem is a main challenge in this field, which limits the wide application of EEG-based emotion recognition. In this article, we propose a novel semisupervised transfer learning framework (EEGMatch) to leverage both

labeled and unlabeled EEG data. First, an EEG-Mixup-based data augmentation method is developed to generate more valid samples for model learning. Second, a semisupervised two-step pairwise learning method is proposed to bridge prototypewise and instancewise pairwise learning, where the prototypewise pairwise learning measures the global relationship between EEG data and the prototypical representation of each emotion class and the instancewise pairwise learning captures the local intrinsic relationship among EEG data. Third, a semisupervised multidomain adaptation is introduced to align the data representation among multiple domains (labeled source domain, unlabeled source domain, and target domain), where the distribution mismatch is alleviated. Extensive experiments are conducted on three benchmark databases (SEED, SEED-IV, and SEED-V) under a cross-subject leave-one-subject-out cross-validation evaluation protocol. The results show the proposed EEGMatch performs better than the state-of-the-art methods under different incomplete label conditions (with 5.89% improvement on SEED, 0.93% improvement on SEED-IV, and 0.28% improvement on SEED-V), which demonstrates the effectiveness of the proposed EEGMatch in dealing with the label scarcity problem in emotion recognition using EEG signals. The source code is available at <https://github.com/KAZABANA/EEGMatch>.

Received 14 March 2023; revised 28 December 2023, 3 May 2024, and 23 August 2024; accepted 30 October 2024. Date of publication 18 November 2024; date of current version 9 July 2025. This work was supported in part by the National Natural Science Foundation of China under Grant 62276169, Grant 62201356, Grant 22322816, and Grant 82272114; in part by STI 2030-Major Projects under Grant 2021ZD0200500; in part by Shenzhen University–Lingnan University Joint Research Programme; in part by Shenzhen–Hong Kong Institute of Brain Science–Shenzhen Fundamental Research Institutions under Grant 2023SHIBS0003; in part by Shenzhen Science and Technology Research and Development Fund for Sustainable Development Project under Grant KCXFZ20201221173613036; in part by the Medical-Engineering Interdisciplinary Research Foundation of Shenzhen University under Grant 2024YG008; in part by the City University of Hong Kong Project under Grant 9610640; and in part by the InnoHK Projects under Hong Kong Center for Cerebro-Cardiovascular Health Engineering (COCHE). (Rushuang Zhou, Weishan Ye, and Zhiguo Zhang contributed equally to this work.) (Corresponding author: Zhen Liang.)

This work involved human subjects or animals in its research. Approval of all ethical and experimental procedures and protocols was granted by Shanghai Jiao Tong University.

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This article has supplementary downloadable material available at <https://doi.org/10.1109/TNNLS.2024.3493425>, provided by the authors.

Digital Object Identifier 10.1109/TNNLS.2024.3493425

Index Terms—Electroencephalography (EEG), emotion recognition, multidomain adaptation, semisupervised learning, transfer learning.

I. INTRODUCTION

EMOTION recognition using electroencephalography (EEG) signals is a rapid and exciting growing field that recently has attracted considerable interest in affective computing and brain–computer interface [1]. The majority of the existing EEG-based emotion recognition studies are based on traditional supervised learning, in which all the data in the training set are well labeled. Unfortunately, labeled data are typically difficult, expensive, and time-consuming to obtain, requiring a tremendous human effort. On the other hand, unlabeled data are easier to collect in contrast to labeled data. Compared with supervised learning, semisupervised learning, which uses both labeled and unlabeled data, could involve less human effort, solve the modeling issue with small labeled training data, and create models that are more generalizable. However, how to design algorithms that efficiently and effectively leverage the presence of both labeled and unlabeled data for improving model learning is now a crucial challenge in semisupervised EEG-based emotion recognition.

In the existing supervised EEG-based emotion recognition studies, transfer learning methods have been widely adopted to minimize the individual differences in EEG signals [2], [3], [4], [5], [6], where the labeled data from the training subjects are considered as the **source domain** and the unknown data from the testing subjects are treated as the **target domain**. By aligning the joint distribution of the source and target domains for approximately satisfying the independent and identically distributed (IID) assumption, the individual differences in EEG signals could be alleviated, and the model performance on the target domain could be enhanced [2], [7]. More introductions about the existing EEG-based emotion recognition studies with transfer learning methods are presented in Section II.

In order to achieve reliable model training, the current EEG-based emotion recognition studies using transfer learning methods often need a substantial amount of labeled data in the source domain. If a majority number of data in the source domain are unlabeled, the model training performance would be severely constrained. Semisupervised learning framework provides great potential to address this problem by taking advantage of a combination of a small number of labeled data and a large amount of unlabeled data and enhancing learning tasks. For example, Zhang and Etemad [8] utilized a holistic semisupervised learning method [9] for EEG representation and showed the superiority of the semisupervised learning framework when the labeled data are scarce. Furthermore, Zhang et al. [10] combined the holistic semisupervised learning approaches and domain adaptation methods to improve the model performance on cross-subject emotion recognition. Still, there are three limitations in the literature, which should be thoroughly investigated and resolved in future studies on semisupervised EEG-based cross-subject emotion recognition.

- 1) *Inappropriate Data Augmentation for EEG Signals:* Previous studies utilized the Mixup method [11] for EEG data augmentation, which ignored the nonstationarity property [12] of EEG signals. The EEG samples collected from different subjects at different trials are not IID, which contradicts the main tenet of the Mixup method.
- 2) *Inefficient Learning Framework for Incomplete Label Task:* Previous works were based on a pointwise learning framework that relied on abundant and precise labels for model training and were impractical under label scarcity conditions [13].
- 3) *Ignoring the Distribution Mismatch Between Labeled and Unlabeled Source Data:* Previous models only considered the distribution mismatch between the labeled source data and the unknown target data [10], [14]. The distribution alignment for the unlabeled source data was disregarded.

To address the aforementioned issues, we propose a novel semisupervised learning framework (EEGMatch) for EEG-based cross-subject emotion recognition with incomplete labels. The proposed EEGMatch is composed of three main modules: EEG-Mixup-based data augmentation,

semisupervised two-step pairwise learning (prototypewise and instancewise), and semisupervised multidomain adaptation. The main contributions and novelties of this study are listed in the following.

- 1) *Appropriate EEG Data Augmentation Strategy Is Developed:* On the basis of the defined EEG samples that meet the IID assumption, an EEG-Mixup-based data augmentation method is developed to generate proper augmented samples in both the source and target domains, establishing an effective augmentation pipeline for EEG signals.
- 2) *Efficient Learning Framework With Incomplete Labels Is Proposed:* A semisupervised two-step pairwise learning framework, including prototypewise and instancewise, is proposed for proper feature learning with incomplete labels. Here, the semisupervised prototypewise pairwise learning explores global prototypical representation for each emotion category in a semisupervised manner, and the semisupervised instancewise pairwise learning captures the local intrinsic structures of EEG samples, improving the model performance in label scarcity conditions.
- 3) *Distribution Alignment Between Labeled and Unlabeled Source Data Is Incorporated:* A semisupervised multidomain adaptation method is designed to jointly minimize the distribution shift among the labeled source data, unlabeled source data, and unknown target data. It could significantly enhance model generalization, as demonstrated by the derived error bound.

II. RELATED WORK

A. EEG-Based Emotion Recognition

In the past ten years, the feasibility and stability of using EEG signals to identify individuals' emotional states have been widely explored [15], [16], [17], [18], [19], [20], [21], [22], [23]. In 2013, Duan et al. [15] extracted EEG features to represent the characteristics associated with emotional states and used the support vector machine (SVM) for emotion classification. With the rapid advancement of deep learning technologies, more and more EEG-based emotion recognition studies have been developed based on deep neural network structures. For example, Zhang et al. [19] proposed cascade and parallel convolutional recurrent neural networks to learn effective emotion-related EEG patterns. Song et al. [24] introduced dynamical graph convolutional neural networks to model the multichannel EEG features and dynamically capture the intrinsic relationship among different channels. To extract dynamic multilevel spatial information among EEG channels, Ye et al. [25] proposed a hierarchical dynamic graph convolutional network (HD-GCN). Li et al. [26] developed a multiple emotion-related spatial model to extract discriminative graph topologies in EEG brain networks and perform efficient emotion recognition. Niu et al. [27] developed a novel deep residual neural network with a combination of brain network analysis and channel-spatial attention mechanism, which enabled an effective classification of multiple emotion states. Aiming at incorporating the spatial and temporal information

from EEG signals, Cheng et al. [28] developed a hybrid network (HN-DGTS) by combining temporal self-attention and dynamic graph convolution. The above models are robust in subject-dependent (within-subject) emotion recognition, where the training and testing data are sampled from the same subject. However, due to the individual differences in EEG signals, the model performance would deteriorate on the subject-independent (cross-subject) emotion-recognition tasks, where the training and testing data are sampled from different subjects [29], [30], [31].

B. EEG-Based Cross-Subject Emotion Recognition With Transfer Learning Methods

To tackle the individual differences in the EEG signals, the existing studies have introduced transfer learning methods to eliminate the feature distribution discrepancy extracted from different subjects and improve the model stability on EEG-based cross-subject emotion recognition tasks [4], [5], [6], [29], [32], [33], [34]. Based on nondeep transfer learning methods such as transfer component analysis (TCA) and transductive parameter transfer (TPT), Zheng and Lu [29] developed personalized emotion recognition models to improve the cross-subject model performance, where the proposed transfer learning framework achieved a mean accuracy of 76.31% (the obtained mean accuracy was only 56.73% when a traditional nontransfer learning framework was adopted). Based on deep transfer learning methods, Jin et al. [32] proposed an EEG-based emotion recognition model using the domain adversarial neural network (DANN) [35]. It was found that the models using the deep transfer learning framework outperformed those using the nondeep transfer learning methods, with the mean accuracy rising from 76.31% to 79.19%. Furthermore, a series of enhanced deep transfer learning frameworks based on the DANN structure has been developed for cross-subject emotion recognition. For example, Li et al. [33] proposed a domain adaptation method to minimize the distribution shift and generalize the emotion recognition models across subjects. He et al. [6] combined the adversarial discriminator and the temporal convolutional networks (TCNs) to further enhance the distribution matching in DANN. Huang et al. [5] developed a generator-based domain adaptation model with a knowledge-free mechanism to maintain the neurological information during the feature alignment. Xu et al. [4] utilized the graph attention adversarial training and biological topology information to construct the domain adversarial graph attention model (DAGAM). Peng et al. [36] proposed a joint feature adaptation and graph adaptive label propagation method (JAGP) for performance enhancement. In comparison with the original DANN structure, these models achieved superior cross-subject emotion recognition results. In order to build reliable models, each of the aforementioned models needs a significant amount of labeled EEG data in the source domain. However, gathering enough good-quality labeled EEG data is always challenging and time-consuming. It is important to develop new transfer learning strategies for efficient cross-subject emotion recognition from a few labeled data and sufficient unlabeled data. This new setting can be

named as semisupervised EEG-based cross-subject emotion recognition.

C. Semisupervised EEG-Based Cross-Subject Emotion Recognition

In contrast to prior studies of supervised EEG-based cross-subject emotion recognition, Dan et al. [37] proposed a possibilistic clustering-promoting semisupervised model for EEG-based emotion recognition, with less requirement on the labeled training data. Due to the less consideration of the distribution shift between the source and target domains, the cross-subject emotion recognition performance was constrained. To reduce the distribution mismatch between the source and target domains, Zhang et al. [10] proposed a semisupervised EEG-based emotion recognition model (PARSE) based on DANN and pairwise representation alignment. The experimental results showed that the proposed method can greatly improve the semisupervised EEG-based cross-subject emotion recognition performance.

However, there are two main limitations of the aforementioned semisupervised EEG-based cross-subject emotion recognition studies.

- 1) *Inappropriate Selection of Labeled and Unlabeled Data:* The previous studies randomly selected the labeled and unlabeled data in terms of the 1-s segments instead of trials. It is noted that in the benchmark databases (SEED [16] and SEED-IV [38]), one trial was composed of a number of 1-s segments, and the same emotional label was assigned to all of the 1-s segments from the same trial. Therefore, a random selection of labeled and unlabeled data in terms of 1-s segments would lead to information leaking and improper model evaluation. Considering the data continuity, we suggest the selection of labeled and unlabeled data in terms of trials to remove the effect of random seeds. Specifically, for each subject in the training set, we select the first N trials as the labeled source domain ($\mathbb{S}$) (both EEG data and labels are available) and the remaining trials as the unlabeled source domain ($\mathbb{U}$) (only EEG data are available). For the subject(s) in the testing set [the target domain ($\mathbb{T}$)], only EEG data are available.
- 2) *Inappropriate Usage of Unlabeled Source Data:* Both Dan et al. [37] and Zhang et al. [10] did not use the unlabeled source data for model training, which led to a significant loss of data information. Using both labeled and unlabeled source data for modeling would encourage the model to perform more reliably and generalize better to unknown data. In this study, we propose to jointly utilize both labeled and unlabeled source data for model training and develop a semisupervised multidomain adaption method to address the distribution alignment issue among $\mathbb{S}$, $\mathbb{U}$, and $\mathbb{T}$.

III. METHODOLOGY

Suppose that the EEG samples in $\mathbb{S}$, $\mathbb{U}$, and $\mathbb{T}$ are given as $D_s = \{X_s, Y_s\}$, $D_u = \{X_u, Y_u\}$, and $D_t = \{X_t, Y_t\}$, respectively. Here, $\{X_s, Y_s\} = \{(x_i^s, y_i^s)\}_{i=1}^{N_s}$,

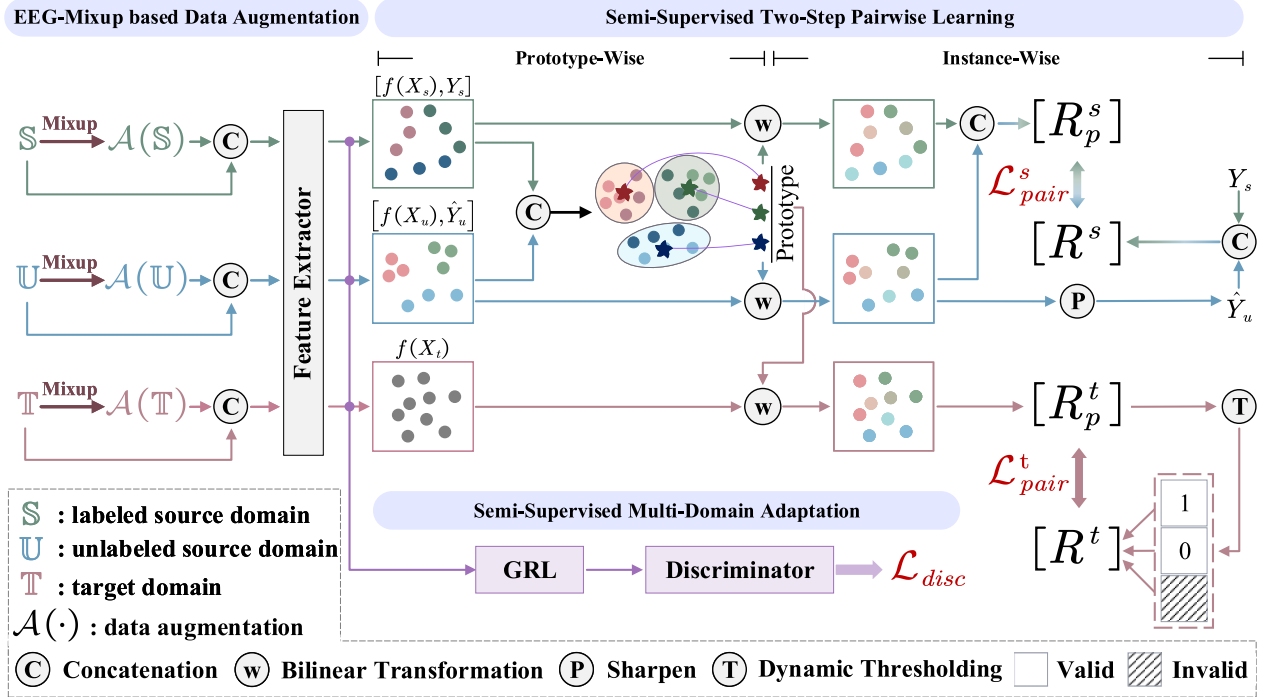


Fig. 1. Semisupervised learning framework of the proposed EEGMatch. It consists of three modules: 1) EEG-Mixup-based data augmentation: generate augment data and increase the sample size for modeling; 2) semisupervised two-step pairwise learning: boost the feature learning under consideration of global feature representation (prototypewise) and local intrinsic relationship (instancewise); and 3) semisupervised multidomain adaptation: align the distribution shift among the labeled source domain ($\mathbb{S}$), the unlabeled source domain ($\mathbb{U}$), and the target domain ($\mathbb{T}$). Here, $\mathcal{L}_{pair}^s$, $\mathcal{L}_{pair}^t$, and $\mathcal{L}_{disc}$ are the semisupervised pairwise learning loss in the source domain, the unsupervised pairwise learning loss in the target domain, and the domain discriminator loss, defined in (14), (17), and (25), respectively.

$\{X_u, Y_u\} = \{(x_i^u, y_i^u)\}_{i=1}^{N_u}$, and $\{X_t, Y_t\} = \{(x_i^t, y_i^t)\}_{i=1}^{N_t}$, where x_i^s , x_i^u , and x_i^t are EEG samples and y_i^s , y_i^u , and y_i^t are the corresponding emotion labels. It is noted that y_i^u and y_i^t are not available during model training. As shown in Fig. 1, the proposed model includes three main modules: EEG-Mixup-based data augmentation, semisupervised two-step pairwise learning (prototypewise and instancewise), and semisupervised multidomain adaptation, with three loss functions (semisupervised pairwise learning loss in the source domain, unsupervised pairwise learning loss in the target domain, and multidomain adversarial loss). In the EEG-Mixup-based data augmentation, the augmented data, $\mathcal{A}(\mathbb{S})$, $\mathcal{A}(\mathbb{U})$, and $\mathcal{A}(\mathbb{T})$, are generated in the three domains ($\mathbb{S}$, $\mathbb{U}$, and $\mathbb{T}$) separately. In the semisupervised two-step pairwise learning, prototypewise pairwise learning and instancewise pairwise learning are involved. For the prototypewise pairwise learning, the prototypical representation of each emotion category is learned based on $\mathbb{S}$ and $\mathbb{U}$, and the pairwise relationships between all EEG samples and the prototypical representation of different emotion categories are measured. For the instancewise pairwise learning, considering that Y_u and Y_t are unknown during model training, a semisupervised pairwise learning method is developed to reconstruct the pairwise relationship within $\mathbb{S}$ and $\mathbb{U}$, and an unsupervised pairwise learning method is adopted to reconstruct the pairwise relationships in $\mathbb{T}$. In the semisupervised multidomain adaptation, the characterized EEG features in the three domains $[f(X_s), f(X_u),$ and

$f(X_t)]$ are trained to be as indistinguishable as possible, and the distribution shift among the three domains is aligned. More details about each module in the proposed EEGMatch are presented in the following.

A. EEG-Mixup-Based Data Augmentation

The traditional Mixup method [9] has achieved great success in the computer vision field, which requires that the pair of samples used for data augmentation should follow the joint distribution [11]. However, in the EEG-based emotion recognition tasks, the EEG signals collected from different subjects at different trials do not satisfy the IID assumption. In other words, a direct application of the Mixup method on the emotional EEG signals might not be appropriate.

To tackle the data augmentation problem on the EEG signals, we develop a novel EEG-Mixup method to fit the data in accordance with the IID assumption. The proposed EEG-Mixup method is mainly composed of two parts.

- 1) *Valid Data Sampling*: Sampling EEG data that follow the joint distribution. Based on the prior findings [24], [29], [39] that the EEG data from the same trial of the same subject follow the IID assumption, the valid EEG data for data augmentation are defined as the emotional EEG signals from the same subject at the same trial, without overlapping. Take the labeled source domain ($\mathbb{S}$) as an example. The valid data sampling could be represented as $D_{p_s, q_s}^s = \{(x_n^s, y_n^s)\}_{n=1}^{N_{p_s, q_s}^s}$, where N_{p_s, q_s}^s is the total sample size of p_s subject at q_s trial.

2) *EEG Data Augmentation*: Generating augmented data for the three domains. We conduct data augmentation on the emotional EEG signals that are randomly selected from the valid data. For example, (x_i^s, y_i^s) and (x_j^s, y_j^s) are the two data pairs randomly selected from D_{p_s, q_s}^s . The corresponding augmented data (x_z^s, y_z^s) are computed as

$$x_z^s = \omega x_i^s + (1 - \omega)x_j^s \quad (1)$$

$$y_z^s = \omega y_i^s + (1 - \omega)y_j^s \quad (2)$$

where $\omega \sim \text{Beta}(\alpha, \alpha)$ and α is a shape parameter of the Beta distribution. In most of the previous studies [10], [16], [38], [39], [40], [41], the EEG samples from the same trial share the same emotional label ($y_z^s = y_i^s = y_j^s$). Then, the augmented data of D_{p_s, q_s}^s could be termed as

$$\mathcal{A}(D_{p_s, q_s}^s) = \{(x_z^s, y_z^s)\}_{z=1}^{Z_{p_s, q_s}^s} \quad (3)$$

where Z_{p_s, q_s}^s is the number of augmented samples in $\mathcal{A}(D_{p_s, q_s}^s)$. Similarly, the EEG-Mixup-based data augmentation is also conducted in the other two domains ($\mathbb{U}$ and $\mathbb{T}$), and the corresponding augmented data are denoted as $\mathcal{A}(D_{p_u, q_u}^u)$ and $\mathcal{A}(D_{p_t, q_t}^t)$. It is noted that the emotional labels (Y_u and Y_t) in the $\mathbb{U}$ and $\mathbb{T}$ domains are unknown during the training process, and the data augmentation for $\mathbb{U}$ and $\mathbb{T}$ domains is only applied on the EEG data (X_u and X_t), as

$$\mathcal{A}(D_{p_u, q_u}^u) = \{(x_z^u)\}_{z=1}^{Z_{p_u, q_u}^u} \quad (4)$$

$$\mathcal{A}(D_{p_t, q_t}^t) = \{(x_z^t)\}_{z=1}^{Z_{p_t, q_t}^t} \quad (5)$$

where Z_{p_u, q_u}^u and Z_{p_t, q_t}^t are the numbers of augmented samples in $\mathcal{A}(D_{p_u, q_u}^u)$ and $\mathcal{A}(D_{p_t, q_t}^t)$, respectively. For simplicity, $\mathcal{A}(D_{p_s, q_s}^s)$, $\mathcal{A}(D_{p_u, q_u}^u)$, and $\mathcal{A}(D_{p_t, q_t}^t)$ are also represented as $\mathcal{A}(\mathbb{S})$, $\mathcal{A}(\mathbb{U})$, and $\mathcal{A}(\mathbb{T})$ in the following. The final obtained data for the following data processing are a concatenation of the original data and the augmented data in each domain, given as $\{\mathbb{S}, \mathcal{A}(\mathbb{S})\}$, $\{\mathbb{U}, \mathcal{A}(\mathbb{U})\}$, and $\{\mathbb{T}, \mathcal{A}(\mathbb{T})\}$.

B. Semisupervised Two-Step Pairwise Learning

A semisupervised two-step pairwise learning framework is proposed to enhance the feature cohesiveness of EEG samples within the same emotion class and enlarge the feature separability of EEG samples between different emotion classes.

1) *Semisupervised Prototypewise Pairwise Learning*: The first step is conducted in a prototypewise manner. The superiority of prototypical representation in feature learning has been evidenced in various research fields [42], [43], [44], [45]. Specifically, a prototypical representation is an embedding that encodes the most representative information for a given class or relation, serving as the centroid of the features associated with the class [34], [46]. The existing prototypical learning method is conducted in a supervised manner, which incorporates the label information for prototypical representation estimation. For example, for total c emotion classes, the supervised prototypical representation P is computed as

$$P = (Y^T Y)^{-1} Y^T f(X) \quad (6)$$

where $X = [x_1; x_2; \dots; x_n]$ and $Y = [y_1; y_2; \dots; y_n]$ are the EEG data and the corresponding emotional labels (one-hot representation) in the current minibatch, respectively. $f(\cdot)$ is a feature extractor with the parameter θ_f , and $f(X)$ is an $N_B \times M$ feature matrix (N_B is the batch size for stochastic gradient descent and M is the feature dimensionality). $P = [\mu_1; \dots; \mu_i; \dots; \mu_c]$, and μ_i is the prototypical representation of the i th emotion class. For *proof*, see Appendix A in Supplementary Materials.

During the supervised prototypical learning, only the labeled samples in the source domain would be considered in the calculation of P . However, the sample size of the labeled source data could be very limited in most cases. To make full use of both labeled and unlabeled data in the source domain for prototypical learning, we propose a novel semisupervised prototypical learning method by generating the corresponding pseudolabels for the unlabeled data in the source domain as

$$\hat{Y}_u = \text{Sharpen}(\text{Cls}(f(X_u))) \quad (7)$$

where $\text{Cls}(f(X_u))$ is the output prediction of the emotion classifier $\text{Cls}(\cdot)$. Considering the superiority of pairwise learning against traditional pointwise learning [10], [34], [47], [48], the emotion classifier $\text{Cls}(\cdot)$ is defined to estimate the pairwise similarity as

$$l_i = w(f(x^u), \mu_i) = f(x^u)^T B \mu_i, \quad i = 1, \dots, c \quad (8)$$

where $w(\cdot)$ is a bilinear transformation and x^u is an EEG sample from X_u . $B \in R^{d \times d}$ is a trainable bilinear transformation matrix and is not restricted by symmetry or positive definiteness. l_i is the calculated similarity between the data feature $f(x^u)$ and the i th prototypical representation (μ_i). Then, the predicted pseudolabel for x^u could be estimated as

$$\begin{aligned} \text{Cls}(f(x^u)) &= \text{softmax}([l_1, \dots, l_i, \dots, l_c]) \\ &= \text{softmax}(f(x^u)^T B P_{s^*}). \end{aligned} \quad (9)$$

Furthermore, to minimize the entropy in the predicted pseudolabel, we adopt a Sharpen function [9] to adjust the “temperature” of the estimated class distribution $\text{Cls}(f(x^u))$ as

$$\text{Sharpen}(\text{Cls}(f(x^u)), \eta)_c = l_c^{\frac{1}{\eta}} / \sum_{j=1}^c l_j^{\frac{1}{\eta}} \quad (10)$$

where η is a hyperparameter for “temperature” controlling. Then, based on the labeled source data $\{X_s, Y_s\}$ and the unlabeled source data $\{X_u, \hat{Y}_u\}$, (6) could be rewritten as

$$P_{s^*} = \left([Y_s; \hat{Y}_u]^T [Y_s; \hat{Y}_u] \right)^{-1} [Y_s; \hat{Y}_u]^T f([X_s; X_u]). \quad (11)$$

In the implementation, for each minibatch, $\{X_s, Y_s\}$ and $\{X_u, -\}$ are randomly sampled from $\{\mathbb{S}, \mathcal{A}(\mathbb{S})\}$ and $\{\mathbb{U}, \mathcal{A}(\mathbb{U})\}$ defined in Section III-A. $\hat{Y}_u$ is the estimated pseudolabels calculated in (7). Before model training, the prototype matrix P_{s^*} is initialized using (6) with only labeled source data. Note that the proposed method [see (11)] can be considered as a special case of supervised prototypical learning [see (6)]. The main difference is that the proposed method preserves the unlabeled data for prototype computation, while the supervised

method excludes them. The samples in the target domain ($\{\mathbb{T}, \mathcal{A}(\mathbb{T})\}$) are not used in the prototypical learning to avoid overfitting.

2) *Semisupervised Instancewise Pairwise Learning*: The second step is conducted in an instancewise manner. To capture the inherent relationships among the EEG data, we estimate the pairwise similarity among samples.

The instance-to-instance similarity matrix R_p^s in the source domain is calculated as

$$R_p^s(i, j) = \frac{\text{Cls}(f(x_i^{s*}))\text{Cls}(f(x_j^{s*}))^T}{\|\text{Cls}(f(x_i^{s*}))\| \|\text{Cls}(f(x_j^{s*}))\|} \quad (12)$$

where x_i^{s*} and x_j^{s*} are the EEG data sampled from $X^{s*} = [X_s; X_u]$. $X_s \in \{\mathbb{S}, \mathcal{A}(\mathbb{S})\}$ and $X_u \in \{\mathbb{U}, \mathcal{A}(\mathbb{U})\}$. $\text{Cls}(f(\cdot))$ is defined in (9). For the labeled source data X_s , the ground truth is given as Y_s . For the unlabeled data X_u in the source domain, the estimated pseudolabels are $\hat{Y}_u$ given in (7). Then, the ground-truth similarity matrix R^s in the source domain is calculated as

$$R^s = N(Y^{s*})N(Y^{s*})^T \quad (13)$$

where $Y^{s*} = [Y_s; \hat{Y}_u]$ and $N(\cdot)$ is a normalization function that normalizes the column vectors of a given matrix to unit vectors. Then, the instancewise pairwise learning loss in the source domain is defined as

$$\mathcal{L}_{\text{pair}}^s(\theta_f, B) = \frac{1}{N^2} \sum L^s \quad (14)$$

where $L^s = -R^s \log R_p^s - (1 - R^s) \log(1 - R_p^s)$. N indicates the total sample size of the labeled and unlabeled data in the source domain.

For the target domain, the instance-to-instance similarity matrix R_p^t is estimated as

$$R_p^t(i, j) = \frac{\text{Cls}(f(x_i^t))\text{Cls}(f(x_j^t))^T}{\|\text{Cls}(f(x_i^t))\| \|\text{Cls}(f(x_j^t))\|} \quad (15)$$

where x_i^t and x_j^t are sampled from the target-domain data X_t . $\text{Cls}(f(\cdot))$ is defined in (9). The corresponding ground-truth similarity matrix R^t is based on a dynamic thresholding method, given as

$$R^t(i, j) = \begin{cases} 1, & \text{if } R_p^t(i, j) \geq \tau_h \\ 0, & \text{if } R_p^t(i, j) < \tau_l \end{cases} \quad (16)$$

where τ_h and τ_l are the defined dynamic upper and lower thresholds, respectively. Then, the instancewise pairwise learning loss in the target domain is defined as

$$\mathcal{L}_{\text{pair}}^t(\theta_f, B) = \frac{1}{\sum_{i,j} M(i, j)} \sum_{i,j} M(i, j) L^t(i, j) \quad (17)$$

where L^t is a $N_t \times N_t$ pairwise loss matrix, given as $L^t = -R^t \log R_p^t - (1 - R^t) \log(1 - R_p^t)$. M is a $N_t \times N_t$ mask matrix to define the valid pairs involved in the loss calculation, given as

$$M(i, j) = \begin{cases} 1, & \text{if } R^t(i, j) = 1 \text{ or } 0 \\ 0, & \text{if otherwise} \end{cases} \quad (18)$$

where N_t is the total sample size of the unlabeled target-domain samples X_t in the current minibatch. In other words, for the instance-to-instance similarity relationship that falls into the range of $\tau_l \leq R_p^t(i, j) < \tau_h$, the pair relationship (x_i^t, x_j^t) could be considered as invalid and would be excluded in the loss calculation at this round. In order to gradually include more samples for model training, the dynamic thresholds (τ_h and τ_l) are set in a progressive learning strategy, as

$$\begin{cases} \tau_h^t = \tau_h^0 - \frac{\tau_h^0 - \tau_l^0}{2} \times \left(1 - \left(1 - \frac{2}{\text{Maxepoch}}\right)^t\right) \\ \tau_l^t = \tau_l^0 + \frac{\tau_h^0 - \tau_l^0}{2} \times \left(1 - \left(1 - \frac{2}{\text{Maxepoch}}\right)^t\right) \end{cases} \quad (19)$$

where τ_h^t and τ_l^t are the upper and lower thresholds at the current training epoch t , respectively. Maxepoch is the maximum training epoch. τ_h^0 and τ_l^0 are the initial values for the upper and lower thresholds.

Overall, the objective function of the instancewise pairwise learning is a combination of the instancewise pairwise learning loss in the source domain [see (14)] and the instancewise pairwise learning loss in the target domain [see (17)], given as

$$\mathcal{L}_{\text{pair}}(\theta_f, B) = \mathcal{L}_{\text{pair}}^s(\theta_f, B) + \gamma \mathcal{L}_{\text{pair}}^t(\theta_f, B) + \beta \|P_{s^*}^T P_{s^*} - I\|_F \quad (20)$$

where $\|P_{s^*}^T P_{s^*} - I\|_F$ is a penalty term that is applied to avoid redundant feature extraction in the prototypical learning [45] and $\|\cdot\|_F$ is the Frobenius norm. $\gamma = \delta \times (\text{epoch}/\text{Maxepoch})$, where $(\text{epoch}/\text{Maxepoch})$ is the ratio of the current epoch to the maximum epoch. β and δ are the hyperparameters.

C. Semisupervised Multidomain Adaptation

The traditional supervised domain adaptation methods on EEG data were conducted based on DANN [35], where the feature distribution in the source and target domains was aligned [39], [41], [49], [50], [51], [52]. However, in semisupervised learning, treating all the labeled and unlabeled source data as a single domain would increase the difficulty of adaptation [40], [53]. To further eliminate the distribution mismatch among different domains, we introduce a novel semisupervised multidomain adaptation method in this article to align the feature representation among the labeled source domain ($\mathbb{S}$), the unlabeled source domain ($\mathbb{U}$), and the target domain ($\mathbb{T}$). Through the semisupervised multidomain adaptation, the feature distribution discrepancy among the above three domains could be mitigated, and the generalization ability of $f(\cdot)$ and $\text{Cls}(\cdot)$ could be enhanced. The existing multisource multidomain adaptation methods [40], [53], [54] and domain generalization strategies [55] assumed that the label information is complete in the source domain, which is not practical for real-world label scarcity applications. On the contrary, the proposed semisupervised multidomain adaptation method assumes that the label information is incomplete in the source domain to address the limitation of the existing methods.

The theoretical motivation of the proposed semisupervised multidomain adaptation method is first presented. In the existing domain adaptation methods for only two domains (labeled source domain $\mathbb{S}$ and target domain $\mathbb{T}$), the target error $\epsilon_T(h)$ could be bounded as [56]

$$\epsilon_T(h) \leq \delta + \epsilon_S(h) + d_{\mathcal{H}}(\mathcal{D}_S, \mathcal{D}_T) \quad (21)$$

where δ represents the difference in labeling functions across the two domains and is typically assumed to be small under the covariate shift assumption [57]. $\epsilon_S(h)$ is the source error based on a given classification hypothesis h characterized by the source classifier. $d_{\mathcal{H}}(\mathcal{D}_S, \mathcal{D}_T)$ corresponds to the $\mathcal{H}$ -divergence introduced in [58], which measures the divergence between the source-domain distribution $\mathcal{D}_S$ and the target-domain distribution $\mathcal{D}_T$ and could be estimated directly from the error of a binary classifier trained to distinguish the domains [59]. Alternatively, an accurate estimation of the source error and the $\mathcal{H}$ -divergence is important for the optimization of the target error.

However, in semisupervised learning, an accurate estimation of the source error and the $\mathcal{H}$ -divergence would be extremely challenging with only a few labeled source data. To tackle this issue, we suggest using both labeled and unlabeled samples in the source domain for the estimation of the source error and the $\mathcal{H}$ -divergence. In the proposed semisupervised multidomain adaptation method, the source domain is given as $\mathbb{S}^* = \{\mathbb{S}, \mathbb{U}\}$, which is composed of both the labeled source data ($\mathbb{S}$) and the unlabeled source data ($\mathbb{U}$). The convex hull $\Lambda_{S_\pi^*}$ of S_π^* is a set of mixture distributions, given as

$$\Lambda_{S_\pi^*} = \{\bar{\mathcal{D}}_{S_\pi^*} : \bar{\mathcal{D}}_{S_\pi^*}(\cdot) = \pi_S \mathcal{D}_S(\cdot) + \pi_U \mathcal{D}_U(\cdot)\} \quad (22)$$

where $\bar{\mathcal{D}}_{S_\pi^*}$ is a distribution computed by the weighted summation of the labeled source-domain distribution $\mathcal{D}_S$ and the unlabeled source-domain distribution $\mathcal{D}_U$. π_S and π_U are the corresponding weights, belonging to Δ_1 (a first dimensional simplex). For the target domain $\mathbb{T}$, $\bar{\mathcal{D}}_T$ is the closest element to $\mathcal{D}_T$ within $\Lambda_{S_\pi^*}$, which can be calculated as

$$\operatorname{argmin}_{\pi_S, \pi_U} d_{\mathcal{H}}[\mathcal{D}_T, \pi_S \mathcal{D}_S + \pi_U \mathcal{D}_U]. \quad (23)$$

Then, the generalization upper bound for the target error $\epsilon_T(h)$ could be derived using the previous domain adaptation methods [59], [60], [61].

Theorem 1: For a given classification hypothesis h ($\forall h \in \mathcal{H}$), the target error $\epsilon_T(h)$ is bounded as

$$\begin{aligned} \epsilon_T(h) \leq & \epsilon_S(h) + \min\{\mathbb{E}_{\bar{\mathcal{D}}_T}[|F_{\bar{T}} - F_T|], \mathbb{E}_{\mathcal{D}_T}[|F_T - F_{\bar{T}}|]\} \\ & + \pi_U(d_{\mathcal{H}}(\mathcal{D}_S, \mathcal{D}_U) + d_{\mathcal{H}}(\mathcal{D}_U, \mathcal{D}_T)) + \pi_S d_{\mathcal{H}}(\mathcal{D}_S, \mathcal{D}_T) \\ & + \pi_U \min\{\mathbb{E}_{\mathcal{D}_S}[|F_S - F_U|], \mathbb{E}_{\mathcal{D}_U}[|F_U - F_S|]\} \end{aligned} \quad (24)$$

where $\epsilon_S(h)$ is the error of the labeled source domain and $d_{\mathcal{H}}(\cdot)$ represents the $\mathcal{H}$ -divergence between the given domains. $F_{\bar{T}}(x) = \pi_S F_S(x) + \pi_U F_U(x)$ is the labeling function for any $x \in \operatorname{Supp}(\bar{\mathcal{D}}_T)$. $F_T(x)$ is the labeling function of the target domain. Note that the labeling function in this study includes the feature extractor $f(\cdot)$ and emotion classifier $\text{Cls}(\cdot)$.

For *proof*, see Appendix B in Supplementary Materials.

Under an extreme case where all the samples in the source domain are labeled ($\mathbb{U}$ is empty), the bound in Theorem 1

is equal to the bound given in (21). Next, we will show how to optimize the target error empirically in the proposed semisupervised multidomain adaptation method. For an ideal joint hypothesis across $\mathbb{S}$, $\mathbb{U}$, and $\mathbb{T}$ domains, the second term ($\min\{\mathbb{E}_{\bar{\mathcal{D}}_T}[|F_{\bar{T}} - F_T|], \mathbb{E}_{\mathcal{D}_T}[|F_T - F_{\bar{T}}|]\}$) and the last term ($\pi_U \min\{\mathbb{E}_{\mathcal{D}_S}[|F_S - F_U|], \mathbb{E}_{\mathcal{D}_U}[|F_U - F_S|]\}$) in Theorem 1 is assumed to be small under the covariate shift assumption [56], [57], [60], [62].

1) *Empirical Minimization of the $\mathcal{H}$ -Divergence:* Based on Theorem 1, it is found that a minimization of the $\mathcal{H}$ -divergence among the domains $\mathbb{S}$, $\mathbb{U}$ and $\mathbb{T}$ is critical for the optimization of the target error $\epsilon_T(h)$. Previous studies have proved that the empirical $\mathcal{H}$ -divergence can be approximated by the classification error of the domain discriminator [35], [59]. In other words, the minimization of the $\mathcal{H}$ -divergence can be achieved by maximizing the domain discriminator loss via adversarial training. Considering the distribution shift among the given three domains, we redefine the domain discriminator loss as

$$\mathcal{L}_{\text{disc}}(\theta_f, \theta_d) = - \sum_x p(x) \log d(f(x)) \quad (25)$$

where $p(x)$ is a one-hot domain label, referring to the belonging domain of the given EEG sample x . $d(\cdot)$ is a three-class domain discriminator with parameter θ_d , which outputs the domain prediction $d(f(x))$.

2) *Empirical Minimization of the Source Error $\epsilon_S(h)$:* Instead of using the pointwise learning loss (e.g., cross-entropy loss) in the existing studies, we introduce to minimize the instancewise pairwise learning loss, as defined in (20).

3) *Empirical Minimization of the Target Error $\epsilon_T(h)$:* The final objective loss of the proposed EEGMatch is defined as

$$\mathcal{L} = \min_{\theta_f, B} \max_{\theta_d} \mathcal{L}_{\text{pair}}(\theta_f, B) - \lambda \mathcal{L}_{\text{disc}}(\theta_f, \theta_d) \quad (26)$$

where $\mathcal{L}_{\text{pair}}(\theta_f, B)$ is the instancewise pairwise learning loss given in (20) and $\mathcal{L}_{\text{disc}}(\theta_f, \theta_d)$ is the domain discriminator loss defined in (25). λ is a balanced hyperparameter to ensure the stability of adversarial training [35], defined as

$$\lambda = \frac{2}{1 + \exp(-\xi)} - 1 \quad (27)$$

where ξ is a factor related to the training epoch, which is equal to the ratio of the current epoch to the maximum epoch.

The overview of the training details of the EEGMatch is shown in Algorithm S1 (see Supplementary Materials). During the training process of the feature extractor $f(\cdot)$ and the domain discriminator $d(\cdot)$, the model is optimized with a gradient reversal layer (GRL) [35]. The GRL layer can realize the adversarial training by reversing the gradient passed backward from $d(\cdot)$ to $f(\cdot)$. In other words, based on the empirical minimization of the $\mathcal{H}$ -divergence [maximizing the domain discriminator loss $\mathcal{L}_{\text{disc}}(\theta_f, \theta_d)$] and the source error $\epsilon_S(h)$ [minimizing the pairwise loss $\mathcal{L}_{\text{pair}}(\theta_f, B)$], the target error $\epsilon_T(h)$ could be gradually minimized by decreasing the upper bound in the training process, as stated in Theorem 1.

IV. EXPERIMENTAL RESULTS

A. Benchmark Databases

To evaluate the effectiveness of the proposed EEGMatch, we conduct extensive experiments on three well-known public emotional EEG databases: SEED [16], SEED-IV [38], and SEED-V [63]. In the SEED database, a total of three emotions (negative, neutral, and positive) were elicited using 15 movie clips, and the simultaneous EEG signals under the three emotional states were recorded from 15 subjects (seven males and eight females) using the 62-channel ESI Neuroscan system. In the SEED-IV database, a total of four emotions (happiness, sadness, fear, and neutral) were elicited using 24 movie clips, and the simultaneous EEG signals under the four emotional states were also recorded from 15 subjects (seven males and eight females) using the 62-channel ESI Neuroscan system. Each subject attended three separate sessions, and each session included 24 trials corresponding to 24 different movie clips. In the SEED-V database, a total of five emotions (happiness, sadness, fear, disgust, and neutral) were elicited using 15 movie clips, and the simultaneous EEG signals under the five emotions were recorded from 16 subjects (six males and ten females) using the 62-channel ESI Neuroscan system. Each subject attended three separate sessions, and each session included 15 trials corresponding to 15 different movie clips.

To make a fair comparison with the existing studies on the two benchmark databases, we also use the precomputed differential entropy (DE) features as the model input. Specifically, for each trial, the EEG data were divided into a number of 1-s segments, and the DE features were extracted from each 1-s segment at the given five frequency bands (delta, theta, alpha, beta, and gamma) from the 62 channels. Then, for each 1-s segment, it was represented by a 310-D feature vector (5 frequency bands $\times$ 62 channels), which was further filtered by a linear dynamic system method for smooth purpose as the literature [16], [64].

B. Implementation Details and Model Setting

In the model implementation, the feature extractor $f(\cdot)$ and the domain discriminator $d(\cdot)$ are composed of fully connected layers with the ReLU activation function. Specifically, the feature extractor $f(\cdot)$ is designed as 310 neurons (input layer)-64 neurons (hidden layer 1)-ReLU activation-64 neurons (hidden layer 2)-ReLU activation-64 neurons (output feature layer). The domain discriminator $d(\cdot)$ is designed as 64 neurons (input layer)-64 neurons (hidden layer 1)-ReLU activation-dropout layer-64 neurons (hidden layer 2)-3 neurons (output layer)-softmax activation. The size of matrix B given in (8) is 64×64 . For the gradient descent and parameter optimization, we adopt the RMSprop optimizer. The learning rate is set to $1e-3$, and the sample size of the minibatch is 48. All the trainable parameters are randomly initialized from a uniform distribution, and no pretrained models are used. For the hyperparameters, we use $L2$ regularizes ($1e-5$) in the proposed model to avoid overfitting problems. In the EEG-Mixup-based data augmentation, the coefficient α in the beta distribution is set to 0.5, and the ratio of the augmented sample size to the original sample size is set to 1:1. In the semisupervised

two-step pairwise learning, the temperature parameter η used in (10) is set to 0.9 for SEED database and 0.8 for SEED-IV database. The dynamic upper and lower thresholds (τ_h^0 and τ_l^0) used in (19) are initialized as 0.9 and 0.5, respectively. The balance parameter γ used in (20) is controlled by a constant factor δ of 2. The regularization coefficient β used in (20) is set to 0.01. For all the compared deep learning methods reproduced in our experiments, we also use the same set of common hyperparameters and EEG DE features as the proposed EEGMatch to ensure fair comparisons, such as the learning rate and the batch size. For the deep domain adaptation methods DANN [35], DCORAL [65], DDC [66], and DAN [51], we use the same feature extractor $f(\cdot)$ and domain discriminator $d(\cdot)$ as the proposed EEGMatch. The architecture of the classifier used in the methods above is designed as 64 neurons (hidden layer 1)- C neurons (output layer), where C is the number of classes. For the model architecture of the compared deep semisupervised methods MixMatch [9], AdaMatch [14], FlexMatch [67], SoftMatch [68], and PARSE [10], we use the optimal settings introduced in [10]. The model-specific hyperparameters used in different semisupervised methods follow the optimal settings provided by the referenced literature. For the other compared methods, such as TCA [69], we follow the optimal setting and training pipelines introduced in the referenced literature.

C. Experimental Protocol With Incomplete Labels

We also adopt the cross-subject leave-one-subject-out experimental protocol as the existing studies [24], [39], [41], [50] for cross-comparison. Specifically, we treat 14 subjects as the source domain and the remaining 1 subject as the target domain. For the source domain, only the first N trials of each subject in one session have labels (the labeled source domain $\mathbb{S}$), and the rest $N_{\max} - N$ trials of each subject are without labels (the unlabeled source domain $\mathbb{U}$). The remaining subject's N_T trials are considered as the target domain ($\mathbb{T}$) for transfer learning and model evaluation. In the domains $\mathbb{S}$, $\mathbb{U}$, and $\mathbb{T}$, the number of original trials used for EEG-Mixup augmentation is N , $N_{\max} - N$, and N_T , respectively. Given that the augmented sample ratio is set to 1:1, the total number of trials used for model training is $2 \times (N + N_{\max} - N + N_T) = 2 \times (N_{\max} + N_T)$. Assuming that there are Z samples within one original trial, the total number of samples for transfer learning is $2 \times Z \times (N_{\max} + N_T)$. Note that only the original samples in the target domain $\mathbb{T}$ are used for model evaluation. We repeat T times until each subject is treated as the target domain for once, where T is the number of subjects in the database. Then, we compute the averaged accuracy and the standard deviation as the final model performance. It is noted that the label information of $\mathbb{U}$ and $\mathbb{T}$ is unknown during model training. Furthermore, to measure the model stability under different incomplete label conditions, we adjust N at different values. Here, considering that the SEED and SEED-V databases have 15 trials and the SEED-IV database has 24 trials for each subject in one session, the N value varies from 3 to 12, 5 to 10, and 4 to 20 for the SEED, SEED-V, and SEED-IV databases, respectively. To ensure

consistency with a state-of-the-art model in semisupervised emotion recognition [10], we adopt a similar experimental setup, where the first session of the SEED database and three sessions from the SEED-IV and SEED-V databases are used for experiments.

D. Cross-Subject Leave-One-Subject-Out Cross-Validation Results With Incomplete Labels

Table I reports the experimental results on the SEED database, in a comparison with the existing popular machine learning or deep learning methods. The results show that the proposed EEGMatch achieves superior performance compared to the other models under different incomplete label conditions, where the average model performance across $N = 3, 6, 9, 12$ is 91.35 ± 07.03 . Compared to the performance achieved by the best competitor in the literature (DDC [66]: 85.46 ± 08.34), the average performance enhancement under different N values is 5.89%. It demonstrates that even with only a few labeled data (when N is small), EEGMatch still can well utilize both the labeled source data ($\mathbb{S}$) and the unlabeled source data ($\mathbb{U}$) to enhance the model generalization ability and reach a satisfying emotion recognition performance in the target domain ($\mathbb{T}$).

The experimental results on the SEED-IV and SEED-V databases under different incomplete label conditions are reported in Tables II and III, which also show the superiority of the proposed EEGMatch on the semisupervised cross-subject emotion recognition task. Specifically, the average model performance on the SEED-IV database across $N = 4, 8, 12, 16, 20$ is 65.53 ± 08.31 . In addition, the average model performance on the SEED-V database across $N = 5, 15$ is 62.75 ± 09.02 . Compared to the best results in the literature, the average performance enhancement under different N values is 0.93% for the SEED-IV database (DDC [66]: 64.60 ± 08.82) and 0.28% for the SEED-V database (DDC [66]: 62.47 ± 08.25). On the other hand, the improvement of the experimental performance on the two databases is less obvious than that on the SEED database. One possible reason is that more emotion classes in the databases would make the predicted pseudolabels ($\hat{Y}_u$) more unstable.

V. CONCLUSION AND DISCUSSION

To fully analyze the performance and the robustness of the proposed model, we conduct various experiments to evaluate the contribution of each module in the EEGMatch and discuss the effect of different hyperparameter settings. All the present results in this session are based on the SEED database.

A. Ablation Study

In this section, we conduct the ablation study to investigate the contributions of different modules in the proposed model. Table IV reports the cross-subject leave-one-subject-out emotion recognition results on the SEED database under different incomplete label conditions.

- 1) When the EEG-Mixup-based data augmentation module is removed from the framework, the recognition

performance presents a significant decrease for all incomplete label conditions. The average model performance drops from 91.35 ± 07.03 to 89.23 ± 07.76 . It shows that the EEG-Mixup-based data augmentation can benefit the model performance by generating more valid data for model training. We also replace the proposed EEG-Mixup-based data augmentation with the standard Mixup-based data augmentation [11] and find that the standard Mixup-based data augmentation has a negative impact on the performance due to the ignorance of the IID assumption. The average model performance under different N values drops by 3.57%. Note that the number of original trials used for Mixup augmentation is the same as the proposed EEG-Mixup. The augmented sample ratio is also set to 1:1 for both EEG-Mixup and Mixup augmentation methods.

- 2) Comparing the model performance with and without the semisupervised prototypewise pairwise learning, an obvious drop could be observed (from 91.35 ± 07.03 to 88.85 ± 08.89) when the semisupervised prototypewise pairwise learning is removed from the proposed model framework. The benefit of prototypical learning has also been observed in the other few-shot learning or transfer learning studies [34], [44], [45], [79].
- 3) The significant positive impact of the introduced semisupervised instancewise pairwise learning on the model performance is observed, with an average improvement in emotion recognition performance (7.20%) being noted under various N values. The results suggest that the semisupervised instancewise pairwise learning can help the model capture inherent structures among EEG samples and extract representative and informative features for the downstream classification tasks.
- 4) To evaluate the validity of the proposed semisupervised multidomain adaptation method, we replace it using the traditional DANN-based domain adaptation ($\mathbb{U}$ is not considered in the domain adaptation process) and calculate the model performance under different incomplete label conditions. The results show that compared to the traditional DANN-based domain adaptation, the proposed semisupervised multidomain adaptation method (EEGMatch) could estimate a more accurate upper bound for the target error by considering the distribution shift over all multiple domains (labeled source domain $\mathbb{S}$, unlabeled source domain $\mathbb{U}$, and target domain $\mathbb{T}$). On the average, the proposed semisupervised multidomain adaptation method could enhance the model performance by 1.27%.
- 5) Comparing the model performance with and without the progressive learning strategy, a remarkable performance improvement is observed when the strategy is applied. Specifically, the average model performance under different N values increases by 1.45%. Note that we fix the initial upper and lower thresholds throughout the model training process when the updating strategy is removed ($\tau_h^0 = 0.9$ and $\tau_l^0 = 0.5$). Compared with the fixed thresholds, dynamic and progressive thresholds can allow more samples to participate in the training process,

TABLE I

MEAN ACCURACIES (%) AND STANDARD DEVIATIONS (%) OF EMOTION RECOGNITION RESULTS USING CROSS-SUBJECT LEAVE-ONE-SUBJECT-OUT CROSS-VALIDATION ON THE SEED DATABASE WITH INCOMPLETE LABELS. FOR EACH SUBJECT, N TRIALS HAVE LABELS, AND $15 - N$ TRIALS DO NOT HAVE LABELS. THE MODEL RESULTS REPRODUCED BY US ARE INDICATED BY “*”

Methods	$N = 3$	$N = 6$	$N = 9$	$N = 12$	Average
SVM* [70]	65.26±09.67	70.78±08.31	70.61±08.72	68.91±10.56	68.89±09.61
TCA* [69]	58.68±09.66	60.20±09.63	59.93±10.63	60.20±10.59	59.75±10.16
SA* [71]	48.79±10.69	49.68±14.16	54.69±10.35	55.94±08.29	52.27±11.50
KPCA* [72]	57.92±10.36	58.06±08.73	59.39±09.79	60.73±09.36	59.03±09.65
RF* [73]	59.58±11.24	63.14±09.63	67.77±07.83	70.89±07.98	65.35±10.23
Adaboost* [74]	62.45±09.94	68.03±09.78	71.72±08.36	72.59±09.52	68.70±10.23
CORAL* [75]	66.21±09.85	70.16±08.70	73.28±10.30	73.97±10.25	70.90±10.27
GFK* [76]	58.11±11.56	60.12±10.70	60.85±08.94	60.55±10.02	59.91±10.40
KNN* [77]	57.47±11.40	59.87±10.08	60.71±08.75	61.38±09.56	59.86±10.10
DGCNN* [24]	71.75±09.73	79.26±07.00	78.21±07.27	79.11±07.79	77.08±08.60
DAN* [51]	78.57±14.40	84.32±10.30	85.89±08.29	86.31±08.24	83.77±11.05
DANN* [35]	77.41±09.84	83.93±09.42	85.75±07.21	86.74±07.09	83.46±09.23
DCORAL* [65]	74.42±16.02	82.01±09.03	84.00±06.39	83.15±06.98	80.89±11.02
DDC* [66]	80.82±09.37	85.40±09.27	88.06±04.75	87.58±07.00	85.46±08.34
PARSE* [10]	76.34±07.58	79.46±04.63	81.21±04.19	81.73±04.99	79.68±05.90
MixMatch* [9]	63.66±09.85	77.62±05.48	83.31±04.40	81.12±05.17	76.43±10.09
AdaMatch* [14]	74.83±05.71	78.67±08.52	80.90±07.31	81.69±07.03	79.02±07.69
FlexMatch* [67]	76.22±09.09	80.65±06.70	84.04±05.38	84.35±06.41	81.32±07.76
SoftMatch* [78]	74.99±09.33	80.09±06.67	83.84±04.86	83.98±05.66	80.72±07.76
EEGMatch	86.68±07.89	92.70±06.37	93.61±04.03	92.39±06.97	91.35±07.03

TABLE II

MEAN ACCURACIES (%) AND STANDARD DEVIATIONS (%) OF EMOTION RECOGNITION RESULTS USING CROSS-SUBJECT LEAVE-ONE-SUBJECT-OUT CROSS-VALIDATION ON THE SEED-IV DATABASE WITH INCOMPLETE LABELS. FOR EACH SUBJECT, N TRIALS HAVE LABELS, AND $24 - N$ TRIALS DO NOT HAVE LABELS. THE MODEL RESULTS REPRODUCED BY US ARE INDICATED BY “*”

Methods	$N = 4$	$N = 8$	$N = 12$	$N = 16$	$N = 20$	Average
SVM* [70]	45.30±12.26	50.40±14.54	52.59±10.30	47.75±11.34	49.76±12.10	49.16±12.44
TCA* [69]	32.57±07.25	40.63±10.34	42.37±11.42	42.59±10.50	41.82±11.03	40.00±10.89
SA* [71]	29.86±04.90	41.62±12.38	39.45±07.12	35.88±13.08	39.72±10.62	37.31±10.94
KPCA* [72]	35.02±06.92	40.84±09.90	42.18±10.15	43.77±10.24	42.44±12.33	40.85±10.51
RF* [73]	41.51±10.99	45.36±12.58	49.73±09.64	51.42±14.19	49.70±14.56	47.54±13.05
Adaboost* [74]	41.37±09.57	52.37±13.39	52.30±13.51	52.15±13.96	51.96±15.69	50.03±14.06
CORAL* [75]	43.64±11.68	50.62±13.88	52.66±13.97	49.89±13.42	51.17±12.60	49.60±13.50
GFK* [76]	35.30±07.76	44.46±13.03	47.01±12.12	44.44±10.10	41.94±09.73	42.63±11.43
KNN* [77]	35.88±07.67	42.75±11.19	43.51±10.80	41.74±07.48	37.92±09.06	40.36±09.82
DGCNN* [24]	53.69±04.70	60.25±07.68	63.18±09.19	64.07±08.15	62.88±09.55	60.81±08.89
DAN* [51]	55.84±07.37	62.78±08.55	65.59±10.71	64.61±09.04	66.19±07.69	63.00±09.53
DANN* [35]	55.36±08.11	63.74±10.89	66.32±09.77	65.55±09.94	66.20±09.19	63.43±10.48
DCORAL* [65]	56.56±07.97	62.49±09.42	65.65±09.69	64.34±08.38	65.03±08.84	62.81±09.48
DDC* [66]	56.80±07.91	63.88±08.55	67.94±06.60	67.34±07.97	67.03±07.77	64.60±08.82
PARSE* [10]	59.12±07.74	62.34±08.54	65.24±08.88	66.47±08.57	66.82±07.91	64.00±08.83
MixMatch* [9]	57.51±06.16	62.12±05.82	64.11±07.92	65.34±08.39	65.95±08.42	63.01±08.03
AdaMatch* [14]	57.90±08.55	62.67±08.25	64.50±09.35	66.17±08.31	67.44±08.65	63.74±09.25
FlexMatch* [67]	57.59±07.73	60.69±08.50	62.10±09.55	63.24±09.74	63.67±08.93	61.46±09.19
SoftMatch* [78]	58.18±07.70	61.16±08.81	62.15±09.29	62.95±09.39	63.50±08.96	61.59±09.05
EEGMatch	60.58±07.38	65.66±08.75	67.40±09.30	67.75±07.16	66.29±06.55	65.53±08.31

which benefits the model generalization performance [34], [67], [78].

B. Training Performance on Different Domains

The training process of the proposed EEGMatch on different domains is reported in Fig. 2 ($N = 12$). Here, we present the average training performance across different subjects to show the source and target error given in Theorem 1. A higher classification accuracy refers to a lower modeling error. To estimate the $\mathcal{H}$ -divergence among different domains ($\pi_U(d_{\mathcal{H}}(\mathcal{D}_S, \mathcal{D}_U) + d_{\mathcal{H}}(\mathcal{D}_U, \mathcal{D}_T)) + \pi_S d_{\mathcal{H}}(\mathcal{D}_S, \mathcal{D}_T)$), we adopt

the maximum mean discrepancy (MMD) [80] calculation here. The findings reveal that the target error $\epsilon_T(h)$ shown in Fig. 2(c) reduces along with a decline in the source error $\epsilon_S(h)$ shown in Fig. 2(a) and the $\mathcal{H}$ -divergence among different domains shown in Fig. 2(b). We also observe that the average accuracy curve on the target domain exhibits some irregularities during model training. This could be influenced by the nonstationarity of the domain adversarial training [81]. A straightforward way to alleviate this problem is to adjust the learning rate. We carefully examine the impact of varying learning rates on mitigating the observed irregularities and

TABLE III

MEAN ACCURACIES (%) AND STANDARD DEVIATIONS (%) OF EMOTION RECOGNITION RESULTS USING CROSS-SUBJECT LEAVE-ONE-SUBJECT-OUT CROSS-VALIDATION ON THE SEED-V DATABASE WITH INCOMPLETE LABELS. FOR EACH SUBJECT, N TRIALS HAVE LABELS, AND $15 - N$ TRIALS DO NOT HAVE LABELS. THE MODEL RESULTS REPRODUCED BY US ARE INDICATED BY “**”

Methods	$N = 5$	$N = 10$	Average	Methods	$N = 5$	$N = 10$	Average
SVM* [70]	45.46±10.19	45.46±10.85	45.46±10.69	TCA* [69]	30.73±07.96	33.17±08.41	31.95±08.41
SA* [71]	25.90±01.13	30.19±06.06	28.05±04.94	KPCA* [72]	29.13±08.64	32.91±11.44	31.02±10.48
RF* [73]	36.42±10.11	39.42±07.58	37.92±09.21	Adaboost* [74]	40.00±11.82	41.95±09.68	40.98±11.02
CORAL* [75]	44.73±09.89	45.14±10.67	44.94±10.45	GFK* [76]	31.37±07.86	35.63±08.92	33.50±08.81
KNN* [77]	31.61±08.44	34.64±08.70	33.13±08.84	DAN* [51]	54.56±07.05	63.95±09.04	59.25±09.37
DANN* [35]	55.04±07.74	64.01±09.77	59.53±09.89	DCORAL* [65]	56.65±08.83	62.90±08.00	59.78±08.98
DDC* [66]	58.67±07.23	66.26±07.41	62.47±08.25	DGCNN* [24]	45.49±07.17	54.38±11.79	49.93±10.72
MixMatch* [9]	52.83±06.07	63.32±08.93	58.08±09.26	AdaMatch* [14]	58.25±05.19	65.71±07.41	61.98±07.41
FlexMatch* [67]	57.48±11.71	65.06±13.38	61.27±13.13	SoftMatch* [78]	57.74±11.91	64.81±12.55	61.27±12.74
PARSE* [10]	56.00±08.31	65.50±08.18	60.75±09.52	EEGMatch	59.06±07.45	66.44±08.94	62.75±09.02

TABLE IV

ABLATION STUDY OF THE PROPOSED EEGMATCH

Methods	$N = 3$	$N = 6$	$N = 9$	$N = 12$	Average
Without EEG-Mixup based data augmentation	84.93±08.56	89.40±07.81	89.59±06.86	93.01±05.17	89.23±07.76
With standard Mixup based data augmentation	84.93±08.05	88.25±09.38	89.32±07.69	88.59±07.90	87.78±08.45
Without semi-supervised prototype-wise pairwise learning	81.89±11.79	91.45±05.81	90.56±04.53	91.49±07.60	88.85±08.89
Without semi-supervised instance-wise pairwise learning	81.04±07.47	83.09±08.40	85.28±08.16	87.19±08.57	84.15±08.48
With traditional DANN-based domain adaptation	82.69±10.72	93.29±05.54	92.87±06.73	91.48±05.24	90.08±08.56
Without the progressive learning strategy	83.19±10.79	92.01±07.66	92.67±05.50	91.74±06.16	89.90±08.71
EEGMatch	86.68±07.89	92.70±06.37	93.61±04.03	92.39±06.97	91.35±07.03

report the investigation results in Appendix G (see Supplementary Materials).

The learned feature representations by the feature extractor $f(\cdot)$ at different training stages are also visually compared based on the t -distributed stochastic neighbor embedding (t-SNE) algorithm [82]. As shown in Fig. 3(a), it is found that the distribution mismatch of the extracted features from different domains is quite obvious before model training. Along with the minimization of the source error and the $\mathcal{H}$ -divergence defined in Theorem 1, the target error ($\epsilon_T(h)$) could be gradually minimized, as shown in Fig. 3(b) and (c). In addition, the distribution mismatch between different domains is reduced, demonstrating that EEGMatch can address the covariate shift problem caused by the nonstationarity property and the individual differences in EEG signals. Our findings support the effectiveness of the proposed semisupervised multidomain adaptation in EEG modeling under a semisupervised learning framework.

Overall, these results empirically verify the statement in Theorem 1 that the target error $\epsilon_T(h)$ could be gradually minimized by decreasing the upper bound in the training process.

C. Model Performance on Unseen Target Data

In the existing EEG-based emotion recognition models using transfer learning strategy, all the target data are assumed to be complete and available for domain adaptation [6], [10], [29], [32], [39], [41], [51]. However, in real-world applications, modeling with all of the target data is impractical. To further evaluate the stability and reliability of the proposed

EEGMatch on unseen data, we divide the target domain ($\mathbb{T}$) into a validation set ($\mathbb{T}_v$: the first K trials in the target domain) and a test set ($\mathbb{T}_t$: the remaining $15 - K$ in the target domain). Here, the validation set ($\mathbb{T}_v$) is used for domain adaptation, together with the labeled source data ($\mathbb{S}$) and the unlabeled source data ($\mathbb{U}$). Then, the trained model will be validated on the unseen test set ($\mathbb{T}_t$) to evaluate the model generalization capability. It should be noted that $\mathbb{U}$ contains the last three trials of the source subjects in this experiment ($N = 12$). We compare the performance of the EEGMatch with its top four competitors in the SEED database and report the results with different K values in Table V. It can be observed that the proposed EEGMatch achieves the best average performance under different K values, which demonstrates its effectiveness on the unseen data.

D. Benefit of the Introduced Unlabeled Source Domain $\mathbb{U}$

To further investigate the contribution of the introduced unlabeled source domain ($\mathbb{U}$) in solving the semisupervised EEG-based emotion recognition task, we compare the model performance with and without $\mathbb{U}$ under different incomplete label conditions. Note that for the model without $\mathbb{U}$, the data augmentation, two-step pairwise learning, and multidomain adaptation of $\mathbb{U}$ domain are not included. The experimental comparison is presented in Fig. 4. It shows that the introduced unlabeled source domain ($\mathbb{U}$) is beneficial to the emotion recognition task under different label scarcity conditions and helps the model training process to avoid the overfitting problem, especially when the labeled source domain ($\mathbb{S}$) is extremely small.

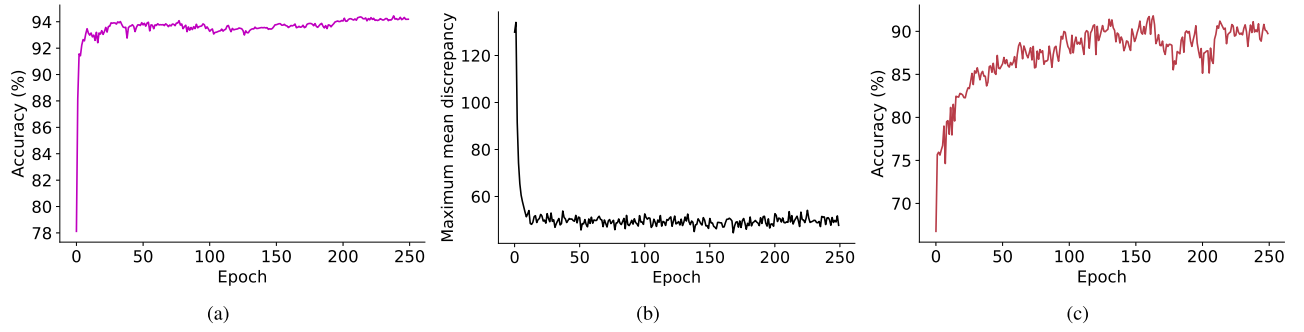


Fig. 2. Training performance on different domains. (a) Training process on the source domain ($\mathcal{S} + \mathcal{U}$). (b) Estimated $\mathcal{H}$ -divergence among different domains in terms of the MMD calculation at different training epochs. (c) Training process on the target domain ($\mathcal{T}$).

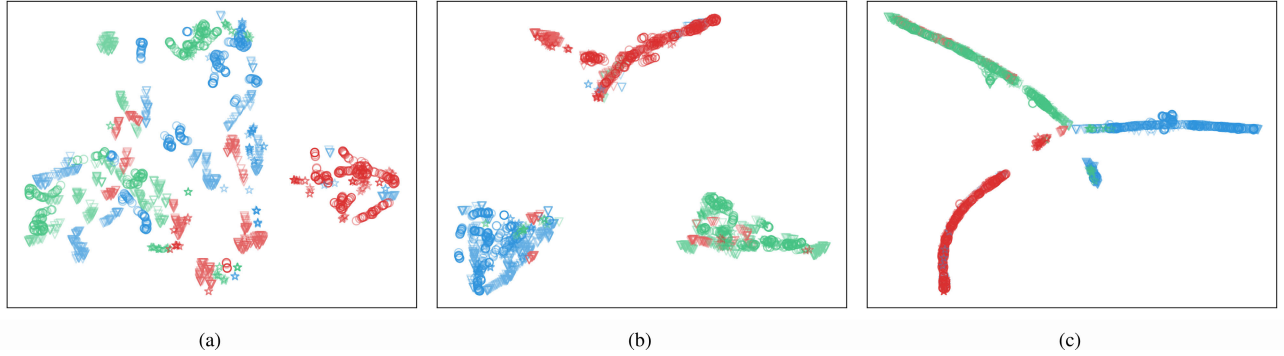


Fig. 3. Visualization of the learned feature representations (a) before training, (b) at the training epoch of 50, and (c) in the final model. Here, the circle, asterisk, and triangle represent the labeled source domain ($\mathcal{S}$), the unlabeled source domain ($\mathcal{U}$), and the target domain ($\mathcal{T}$). The blue, green, and red colors indicate negative, neutral, and positive emotions.

TABLE V
MEAN ACCURACIES (%) AND STANDARD DEVIATIONS (%) OF EMOTION RECOGNITION ON UNSEEN TARGET DATA.
THE MODEL RESULTS REPRODUCED BY US ARE INDICATED BY “**”

Validation Set	Test Set	DAN* [51]	DANN* [35]	DCORAL* [65]	DDC* [66]	EEGMatch
$\mathcal{T}_v = 3$ trials	$\mathcal{T}_t = 12$ trials	81.92 ± 06.35	79.50 ± 06.46	81.02 ± 06.36	82.99 ± 06.98	81.97 ± 07.22
$\mathcal{T}_v = 6$ trials	$\mathcal{T}_t = 9$ trials	81.76 ± 06.25	81.69 ± 08.42	81.16 ± 08.46	82.37 ± 08.81	84.73 ± 06.25
$\mathcal{T}_v = 9$ trials	$\mathcal{T}_t = 6$ trials	82.66 ± 06.26	81.26 ± 07.74	80.48 ± 07.70	82.54 ± 06.85	85.63 ± 06.62
$\mathcal{T}_v = 12$ trials	$\mathcal{T}_t = 3$ trials	84.40 ± 14.23	80.26 ± 14.52	82.23 ± 07.91	84.65 ± 12.47	85.72 ± 10.30
Average Performance		82.69 ± 09.02	80.67 ± 09.83	81.22 ± 07.67	83.14 ± 09.11	84.52 ± 08.34

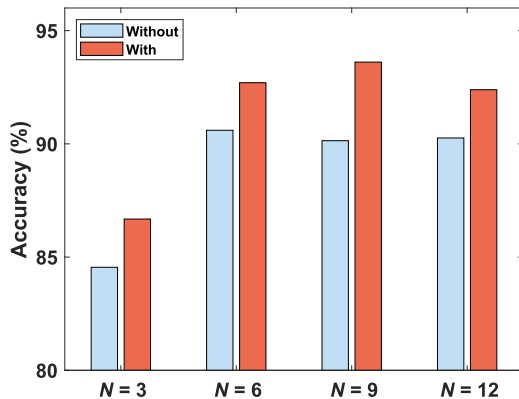


Fig. 4. Emotion recognition accuracies (%) without and with the unlabeled source domain ($\mathcal{U}$) under different N values.

E. Effect of Hyperparameter Settings

The effect of the hyperparameter settings used in the proposed EEGMatch is also examined.

- 1) We change the temperature parameter in the Sharpen function [see (10)] from 0.1 to 1 with a step of 0.1.

The corresponding results are reported in Fig. 5(a). It shows that a change in the temperature parameter would slightly affect the model performance, where the optimal temperature parameter value is 0.2 for the SEED database.

- 2) We adjust the augmentation size (the ratio of the number of augmented samples to the number of original samples) from 0 to 5 in the experiment, where 0 indicates no data augmentation and 5 refers to fourfold data augmentation. As reported in Fig. 5(b), an increase in the augmentation size could improve the model performance; when the augmentation size reaches 2, the model performs at its optimal. When the augmentation size exceeds 4, it is evident that the EEG-Mixup augmentation begins to negatively impact the model's performance. It indicates that a substantial number of the augmented samples might have an adverse effect on the validity and reliability of the training data. This phenomenon has also been reported by other studies [83], [84]. In Appendix D–F (see Supplementary

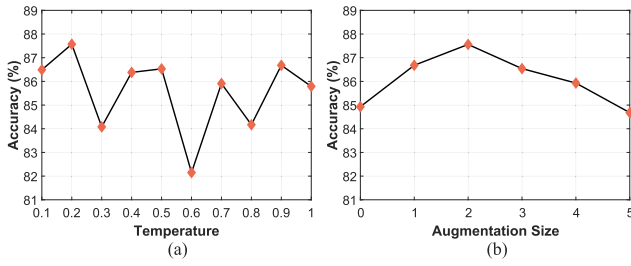


Fig. 5. Emotion recognition accuracies (%) of the proposed EEGMatch at different hyperparameter settings ($N = 3$). (a) Effect of the temperature parameter in the Sharpen function [see (10)]. (b) Effect of the augmentation size in the EEG-Mixup-based data augmentation.

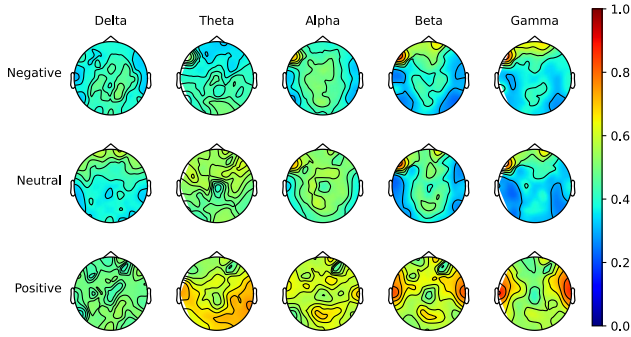


Fig. 6. Topographic analysis of the mutual information between the EEG patterns and the model predictions.

Materials), we provide more sensitivity analysis on the other hyperparameters using the same experiment setting ($N = 3$), such as the number of trainable parameters, the selection of EEG frequency bands, and the initialization of dynamic thresholds.

F. Topographic Analysis of Important EEG Patterns

In the topographic analysis, we identify the important brain patterns for emotion recognition by computing the mutual information between the brain patterns and prediction labels. Specifically, at the i th validation round, we have the data input of the target domain, given as X_i^t , which is an $M \times 310$ matrix. Here, M is the sample size in the target domain, and 310 refers to the extracted DE features from 62 channels at five frequency bands. The corresponding model prediction results are $\hat{Y}_i^t$, with a size of $M \times 3$. The columns in $\hat{Y}_i^t$ indicate the prediction probabilities of different emotions (negative, neutral, and positive). Then, we estimate the mutual information between the input features X_i^t and the prediction results $\hat{Y}_i^t$ using the nonparametric method as stated in [85], [86], and [87]. The obtained mutual information matrix is termed $I(X_i^t, \hat{Y}_i^t) \in R^{3 \times 310}$, indicating a quantification of the inherent dependence between the EEG patterns and the model prediction results. $I(X_i^t, \hat{Y}_i^t)$ is further normalized to $[0, 1]$, and a larger value refers to a greater informativeness of the EEG patterns to the model prediction at the i th validation round. Fig. 6 shows an average of all the obtained $I(X_i^t, \hat{Y}_i^t)$ across different validation rounds. It is found that the EEG patterns with greater informativeness for emotion recognition are mainly located in the beta and gamma frequency bands [16], [24] at the frontal and temporal regions [88], [89], [90], [91].

G. Conclusion

In this article, we present a novel semisupervised learning framework (EEGMatch) for EEG-based cross-subject emotion recognition with incomplete labels. The proposed EEGMatch introduces an efficient EEG-Mixup-based data augmentation to increase the sample size for modeling, develops a semisupervised two-step pairwise learning framework (prototypewise and instancewise) for efficient global and local feature learning, and proposes a semisupervised multidomain adaptation method to jointly tackle the distribution mismatch problem among multiple domains. The extensive experiments are conducted on two benchmark databases (SEED and SEED-IV) under a practical experimental protocol under various label scarcity conditions. In comparison to the existing popular machine learning and deep learning methods, the experimental results demonstrate the superiority of the proposed EEGMatch in tackling semisupervised EEG applications with incomplete labels.

The main performance improvement achieved by the proposed EEGMatch model can be attributed to four factors.

- 1) The proposed EEG-Mixup-based data augmentation method can generate suitable augmented samples for model training, which can help the model avoid the overfitting problem under label scarcity conditions.
- 2) Through the pseudolabel generation with the sharpen function, the proposed semisupervised prototypewise pairwise learning could take advantage of both labeled and unlabeled source data and accurately estimate the prototypical representation of each emotion class.
- 3) The introduced semisupervised instancewise pairwise learning can estimate reliable instance-to-instance relationships and project them into a robust feature embedding space for emotion recognition, where the samples from the same class are pulled together and the samples from different classes are pushed apart.
- 4) The proposed semisupervised multidomain adaptation method can simultaneously align the distribution of different domains and help the model predict accurate labels for the samples in $\mathbb{U}$ and $\mathbb{T}$ domains.

Overall, our work presents an efficient and effective semisupervised learning framework for EEG-based cross-subject emotion recognition. By taking into account the distribution shift, the semisupervised EEG-based cross-subject emotion recognition performance could be greatly enhanced. We believe that the proposed EEGMatch would open up new possibilities for resolving the label missing problem in EEG tasks, in which high-quality labels are challenging to acquire. In the future, further consideration of the class imbalance issue in modeling is necessary, which commonly happens in real-world applications. Moreover, the stabilization of multidomain adversarial training poses an ongoing challenge that warrants increased attention in future research efforts.

REFERENCES

- [1] X. Si, H. He, J. Yu, and D. Ming, "Cross-subject emotion recognition brain-computer interface based on fNIRS and DBNet," *Cyborg Bionic Syst.*, vol. 4, p. 45, Jan. 2023.

- [2] V. Jayaram, M. Alamgir, Y. Altun, B. Scholkopf, and M. Grosse-Wenstrup, "Transfer learning in brain-computer interfaces," *IEEE Comput. Intell. Mag.*, vol. 11, no. 1, pp. 20–31, Feb. 2016.
- [3] Y. Li, W. Zheng, L. Wang, Y. Zong, and Z. Cui, "From regional to global brain: A novel hierarchical spatial-temporal neural network model for EEG emotion recognition," *IEEE Trans. Affect. Comput.*, vol. 13, no. 2, pp. 568–578, Apr. 2022.
- [4] T. Xu, W. Dang, J. Wang, and Y. Zhou, "DAGAM: A domain adversarial graph attention model for subject independent EEG-based emotion recognition," 2022, *arXiv:2202.12948*.
- [5] D. Huang, S. Zhou, and D. Jiang, "Generator-based domain adaptation method with knowledge free for cross-subject EEG emotion recognition," *Cognit. Comput.*, vol. 14, no. 4, pp. 1316–1327, Jul. 2022.
- [6] Z. He, Y. Zhong, and J. Pan, "An adversarial discriminative temporal convolutional network for EEG-based cross-domain emotion recognition," *Comput. Biol. Med.*, vol. 141, Feb. 2022, Art. no. 105048.
- [7] Y. Li, H. Kambara, Y. Koike, and M. Sugiyama, "Application of covariate shift adaptation techniques in Brain-Computer interfaces," *IEEE Trans. Biomed. Eng.*, vol. 57, no. 6, pp. 1318–1324, Jun. 2010.
- [8] G. Zhang and A. Etemad, "Holistic semi-supervised approaches for EEG representation learning," in *Proc. IEEE Int. Conf. Acoust., Speech Signal Process. (ICASSP)*, May 2022, pp. 1241–1245.
- [9] D. Berthelot, N. Carlini, I. Goodfellow, N. Papernot, A. Oliver, and C. Raffel, "MixMatch: A holistic approach to semi-supervised learning," 2019, *arXiv:1905.02249*.
- [10] G. Zhang, V. Davoodnia, and A. Etemad, "PARSE: Pairwise alignment of representations in semi-supervised EEG learning for emotion recognition," 2022, *arXiv:2202.05400*.
- [11] H. Zhang, C. Moustapha, D. Yann N., and L.-P. David, "Mixup: Beyond empirical risk minimization," in *Proc. Int. Conf. Learn. Represent.*, 2018, pp. 1–13.
- [12] V. Rasoulzadeh, E. C. Erkus, T. A. Yogurt, I. Ulusoy, and S. A. Zengeroglu, "A comparative stationarity analysis of EEG signals," *Ann. Oper. Res.*, vol. 258, no. 1, pp. 133–157, Nov. 2017.
- [13] H. Bao, G. Niu, and M. Sugiyama, "Classification from pairwise similarity and unlabeled data," in *Proc. Int. Conf. Mach. Learn.*, 2018, pp. 452–461.
- [14] D. Berthelot, R. Roelofs, K. Sohn, N. Carlini, and A. Kurakin, "AdaMatch: A unified approach to semi-supervised learning and domain adaptation," 2021, *arXiv:2106.04732*.
- [15] R.-N. Duan, J.-Y. Zhu, and B.-L. Lu, "Differential entropy feature for EEG-based emotion classification," in *Proc. IEEE/EMBS Conf. Neural Eng. (NER)*, Aug. 2013, pp. 81–84.
- [16] W.-L. Zheng and B.-L. Lu, "Investigating critical frequency bands and channels for EEG-based emotion recognition with deep neural networks," *IEEE Trans. Auton. Mental Develop.*, vol. 7, no. 3, pp. 162–175, Sep. 2015.
- [17] X. Li, B. Hu, S. Sun, and H. Cai, "EEG-based mild depressive detection using feature selection methods and classifiers," *Comput. Methods Programs Biomed.*, vol. 136, pp. 151–161, Nov. 2016.
- [18] Y.-J. Liu, M. Yu, G. Zhao, J. Song, Y. Ge, and Y. Shi, "Real-time movie-induced discrete emotion recognition from EEG signals," *IEEE Trans. Affect. Comput.*, vol. 9, no. 4, pp. 550–562, Oct. 2018.
- [19] D. Zhang et al., "Cascade and parallel convolutional recurrent neural networks on EEG-based intention recognition for brain computer interface," in *Proc. AAAI Conf. Artif. Intell.*, 2018, vol. 32, no. 1, pp. 1–8.
- [20] F. Yang, X. Zhao, W. Jiang, P. Gao, and G. Liu, "Multi-method fusion of cross-subject emotion recognition based on high-dimensional EEG features," *Frontiers Comput. Neurosci.*, vol. 13, p. 53, Aug. 2019.
- [21] S. K. Khare, V. Bajaj, and G. R. Sinha, "Adaptive tunable q wavelet transform-based emotion identification," *IEEE Trans. Instrum. Meas.*, vol. 69, no. 12, pp. 9609–9617, Dec. 2020.
- [22] Z. Liang et al., "EEGFuseNet: Hybrid unsupervised deep feature characterization and fusion for high-dimensional EEG with an application to emotion recognition," *IEEE Trans. Neural Syst. Rehabil. Eng.*, vol. 29, pp. 1913–1925, 2021.
- [23] L. Jiang et al., "Spatial-rhythmic network as a biomarker of familial risk for psychotic bipolar disorder," *Nature Mental Health*, vol. 1, no. 11, pp. 887–899, 2023.
- [24] T. Song, W. Zheng, P. Song, and Z. Cui, "EEG emotion recognition using dynamical graph convolutional neural networks," *IEEE Trans. Affect. Comput.*, vol. 11, no. 3, pp. 532–541, Sep. 2018.
- [25] M. Ye, C. L. P. Chen, and T. Zhang, "Hierarchical dynamic graph convolutional network with interpretability for EEG-based emotion recognition," *IEEE Trans. Neural Netw. Learn. Syst.*, early access, Dec. 9, 2022, doi: [10.1109/TNNLS.2022.3225855](https://doi.org/10.1109/TNNLS.2022.3225855).
- [26] C. Li et al., "Effective emotion recognition by learning discriminative graph topologies in EEG brain networks," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 35, no. 8, pp. 10258–10272, Aug. 2024.
- [27] W. Niu, C. Ma, X. Sun, M. Li, and Z. Gao, "A brain network analysis-based double way deep neural network for emotion recognition," *IEEE Trans. Neural Syst. Rehabil. Eng.*, vol. 31, pp. 917–925, 2023.
- [28] C. Cheng, Z. Yu, Y. Zhang, and L. Feng, "Hybrid network using dynamic graph convolution and temporal self-attention for EEG-based emotion recognition," *IEEE Trans. Neural Netw. Learn. Syst.*, pp. 1–11, Oct. 2023, doi: [10.1109/TNNLS.2023.3319315](https://doi.org/10.1109/TNNLS.2023.3319315).
- [29] W.-L. Zheng and B.-L. Lu, "Personalizing EEG-based affective models with transfer learning," in *Proc. 25th Int. Joint Conf. Artif. Intell.*, 2016, pp. 2732–2738.
- [30] W. Samek, F. C. Meinecke, and K. Müller, "Transferring subspaces between subjects in brain-computer interfacing," *IEEE Trans. Biomed. Eng.*, vol. 60, no. 8, pp. 2289–2298, Aug. 2013.
- [31] H. Morioka et al., "Learning a common dictionary for subject-transfer decoding with resting calibration," *NeuroImage*, vol. 111, pp. 167–178, May 2015.
- [32] Y.-M. Jin, Y.-D. Luo, W.-L. Zheng, and B.-L. Lu, "EEG-based emotion recognition using domain adaptation network," in *Proc. Int. Conf. Orange Technol. (ICOT)*, Dec. 2017, pp. 222–225.
- [33] J. Li, S. Qiu, C. Du, Y. Wang, and H. He, "Domain adaptation for EEG emotion recognition based on latent representation similarity," *IEEE Trans. Cognit. Develop. Syst.*, vol. 12, no. 2, pp. 344–353, Jun. 2020.
- [34] R. Zhou et al., "PR-PL: A novel prototypical representation based pairwise learning framework for emotion recognition using EEG signals," *IEEE Trans. Affect. Comput.*, vol. 15, no. 2, pp. 657–670, Jun. 2024.
- [35] Y. Ganin et al., "Domain-adversarial training of neural networks," *J. Mach. Learn. Res.*, vol. 17, no. 1, pp. 2030–2096, May 2016.
- [36] Y. Peng, W. Wang, W. Kong, F. Nie, B.-L. Lu, and A. Cichocki, "Joint feature adaptation and graph adaptive label propagation for cross-subject emotion recognition from EEG signals," *IEEE Trans. Affect. Comput.*, vol. 13, no. 4, pp. 1941–1958, Apr. 2022.
- [37] Y. Dan, J. Tao, J. Fu, and D. Zhou, "Possibilistic clustering-promoting semi-supervised learning for EEG-based emotion recognition," *Frontiers Neurosci.*, vol. 15, Jun. 2021, Art. no. 690044.
- [38] W.-L. Zheng, W. Liu, Y. Lu, B.-L. Lu, and A. Cichocki, "EmotionMeter: A multimodal framework for recognizing human emotions," *IEEE Trans. Cybern.*, vol. 49, no. 3, pp. 1110–1122, Mar. 2018.
- [39] X. Du et al., "An efficient LSTM network for emotion recognition from multichannel EEG signals," *IEEE Trans. Affect. Comput.*, vol. 13, no. 3, pp. 1528–1540, Jul. 2022.
- [40] J. Li, S. Qiu, Y.-Y. Shen, C.-L. Liu, and H. He, "Multisource transfer learning for cross-subject EEG emotion recognition," *IEEE Trans. Cybern.*, vol. 50, no. 7, pp. 3281–3293, Jul. 2020.
- [41] P. Zhong, D. Wang, and C. Miao, "EEG-based emotion recognition using regularized graph neural networks," *IEEE Trans. Affect. Comput.*, vol. 13, no. 3, pp. 1290–1301, Jul. 2022.
- [42] K. Wang, J. H. Liew, Y. Zou, D. Zhou, and J. Feng, "PANet: Few-shot image semantic segmentation with prototype alignment," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 9197–9206.
- [43] X. Gu, "A self-training hierarchical prototype-based approach for semi-supervised classification," *Inf. Sci.*, vol. 535, pp. 204–224, Oct. 2020.
- [44] J. Snell, K. Swersky, and R. Zemel, "Prototypical networks for few-shot learning," in *Proc. Adv. Neural Inf. Process. Syst.*, I. Guyon et al., Eds., Jul. 2020, pp. 178–181.
- [45] P. O. Pinheiro, "Unsupervised domain adaptation with similarity learning," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 8004–8013.
- [46] N. Ding et al., "Prototypical representation learning for relation extraction," 2021, *arXiv:2103.11647*.
- [47] H. Bao, T. Shimada, L. Xu, I. Sato, and M. Sugiyama, "Pairwise supervision can provably elicit a decision boundary," in *Proc. 25th Int. Conf. Artif. Intell. Statist.*, in Proceedings of Machine Learning Research, vol. 151, G. Camps-Valls, F. J. R. Ruiz, and I. Valera, Eds., 2022, pp. 2618–2640.
- [48] P. Khosla et al., "Supervised contrastive learning," in *Proc. NIPS*, vol. 33, 2020, pp. 18661–18673.

- [49] Y. Luo, S. Y. Zhang, W. L. Zheng, and B. L. Lu, "WGAN domain adaptation for EEG-based emotion recognition," in *Proc. Int. Conf. Neural Inf. Process.*, 2018, pp. 275–286.
- [50] Y. Li et al., "A novel bi-hemispheric discrepancy model for EEG emotion recognition," *IEEE Trans. Cognit. Develop. Syst.*, vol. 13, no. 2, pp. 354–367, Jun. 2020.
- [51] H. Li, Y.-M. Jin, W.-L. Zheng, and B.-L. Lu, "Cross-subject emotion recognition using deep adaptation networks," in *Proc. Int. Conf. Neural Inf. Process. (ICONIP)*, L. Cheng, A. C. S. Leung, and S. Ozawa, Eds. Cham, Switzerland: Springer, 2018, pp. 403–413.
- [52] L.-M. Zhao, X. Yan, and B.-L. Lu, "Plug-and-play domain adaptation for cross-subject EEG-based emotion recognition," in *Proc. AAAI Conf. Artif. Intell.*, 2021, vol. 35, no. 1, pp. 863–870.
- [53] H. Chen, M. Jin, Z. Li, C. Fan, J. Li, and H. He, "MS-MDA: Multisource marginal distribution adaptation for cross-subject and cross-session EEG emotion recognition," *Frontiers Neurosci.*, vol. 15, Dec. 2021, Art. no. 778488.
- [54] H. Chen, Z. Li, M. Jin, and J. Li, "MEERNet: Multi-source EEG-based emotion recognition network for generalization across subjects and sessions," in *Proc. 43rd Annu. Int. Conf. IEEE Eng. Med. Biol. Soc. (EMBC)*, Nov. 2021, pp. 6094–6097.
- [55] B.-Q. Ma, H. Li, W.-L. Zheng, and B.-L. Lu, "Reducing the subject variability of EEG signals with adversarial domain generalization," in *Proc. Int. Conf. Neural Inf. Process.*, T. Gedeon, K. W. Wong, and M. Lee, Eds. Cham, Switzerland: Springer, 2019, pp. 30–42.
- [56] S. Ben-David, J. Blitzer, K. Crammer, A. Kulesza, F. Pereira, and J. W. Vaughan, "A theory of learning from different domains," *Mach. Learn.*, vol. 79, no. 1, pp. 151–175, 2010.
- [57] S. Ben-David, T. Lu, T. Luu, and D. Pal, "Impossibility theorems for domain adaptation," in *Proc. 13th Int. Conf. Artif. Intell. Statist.*, vol. 9, 2010, pp. 129–136.
- [58] D. Kifer, S. Ben-David, and J. Gehrke, "Detecting change in data streams," in *Proc. VLDB*, vol. 4, Toronto, ON, Canada, 2004, pp. 180–191.
- [59] S. Ben-David, J. Blitzer, K. Crammer, and F. Pereira, "Analysis of representations for domain adaptation," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 19, 2006, pp. 1–8.
- [60] I. Albuquerque, J. Monteiro, M. Darvishi, T. H. Falk, and I. Mitliagkas, "Generalizing to unseen domains via distribution matching," 2019, *arXiv:1911.00804*.
- [61] H. Zhao, S. Zhang, G. Wu, J. M. Moura, J. P. Costeira, and G. J. Gordon, "Adversarial multiple source domain adaptation," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 31, 2018, pp. 1–12.
- [62] A. Sicilia, X. Zhao, and S. J. Hwang, "Domain adversarial neural networks for domain generalization: When it works and how to improve," 2021, *arXiv:2102.03924*.
- [63] W. Liu, J.-L. Qiu, W.-L. Zheng, and B.-L. Lu, "Comparing recognition performance and robustness of multimodal deep learning models for multimodal emotion recognition," *IEEE Trans. Cognit. Develop. Syst.*, vol. 14, no. 2, pp. 715–729, Jun. 2022.
- [64] L.-C. Shi and B.-L. Lu, "Off-line and on-line vigilance estimation based on linear dynamical system and manifold learning," in *Proc. Annu. Int. Conf. IEEE Eng. Med. Biol.*, Aug. 2010, pp. 6587–6590.
- [65] B. Sun and K. Saenko, "Deep CORAL: Correlation alignment for deep domain adaptation," in *Computer Vision—ECCV*. Cham, Switzerland: Springer, 2016, pp. 443–450.
- [66] E. Tzeng, J. Hoffman, N. Zhang, K. Saenko, and T. Darrell, "Deep domain confusion: Maximizing for domain invariance," 2014, *arXiv:1412.3474*.
- [67] B. Zhang et al., "FlexMatch: Boosting semi-supervised learning with curriculum pseudo labeling," in *Proc. NIPS*, 2021, pp. 18408–18419.
- [68] K. Sohn et al., "FixMatch: Simplifying semi-supervised learning with consistency and confidence," in *Proc. Adv. Neural Inf. Process. Syst. (NIPS)*, vol. 33, 2020, pp. 596–608.
- [69] S. J. Pan, I. W. Tsang, J. T. Kwok, and Q. Yang, "Domain adaptation via transfer component analysis," *IEEE Trans. Neural Netw.*, vol. 22, no. 2, pp. 199–210, Feb. 2011.
- [70] J. A. K. Suykens and J. Vandewalle, "Least squares support vector machine classifiers," *Neural Process. Lett.*, vol. 9, no. 3, pp. 293–300, Jun. 1999.
- [71] B. Fernando, A. Habrard, M. Sebban, and T. Tuytelaars, "Unsupervised visual domain adaptation using subspace alignment," in *Proc. IEEE Int. Conf. Comput. Vis.*, Dec. 2013, pp. 2960–2967.
- [72] S. Mika, B. Schölkopf, A. Smola, K.-R. Müller, M. Scholz, and G. Rätsch, "Kernel PCA and de-noising in feature spaces," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 11, 1999, pp. 1–11.
- [73] L. Breiman, "Random forests," *Mach. Learn.*, vol. 45, pp. 5–32, Oct. 2001.
- [74] T. Hastie, S. Rosset, J. Zhu, and H. Zou, "Multi-class AdaBoost," *Statist. Interface*, vol. 2, no. 3, pp. 349–360, 2009.
- [75] B. Sun, J. Feng, and K. Saenko, "Return of frustratingly easy domain adaptation," in *Proc. 30th AAAI Conf. Artif. Intell.*, 2016, pp. 2058–2065.
- [76] B. Gong, Y. Shi, F. Sha, and K. Grauman, "Geodesic flow kernel for unsupervised domain adaptation," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2012, pp. 2066–2073.
- [77] D. Coomans and D. L. Massart, "Alternative k-nearest neighbour rules in supervised pattern recognition: Part 1. K-nearest neighbour classification by using alternative voting rules," *Analytica Chim. Acta*, vol. 136, pp. 15–27, Aug. 1982.
- [78] H. Chen et al., "SoftMatch: Addressing the quantity-quality trade-off in semi-supervised learning," 2023, *arXiv:2301.10921*.
- [79] K. Allen, E. Shelhamer, H. Shin, and J. Tenenbaum, "Infinite mixture prototypes for few-shot learning," in *Proc. Int. Conf. Mach. Learn.*, vol. 97, K. Chaudhuri and R. Salakhutdinov, Eds., Jun. 2019, pp. 232–241.
- [80] D. Sejdinovic, B. Sriperumbudur, A. Gretton, and K. Fukumizu, "Equivalence of distance-based and RKHS-based statistics in hypothesis testing," *Ann. Stat.*, vol. 41, no. 5, pp. 2263–2291, 2013.
- [81] M. Arjovsky, S. Chintala, and L. Bottou, "Wasserstein generative adversarial networks," in *Proc. Int. Conf. Mach. Learn.*, 2017, pp. 214–223.
- [82] L. van der Maaten and G. Hinton, "Visualizing data using t-SNE," *J. Mach. Learn. Res.*, vol. 9, pp. 2579–2605, Nov. 2008.
- [83] Y. Luo and B. Lu, "EEG data augmentation for emotion recognition using a conditional Wasserstein GAN," in *Proc. 40th Annu. Int. Conf. IEEE Eng. Med. Biol. Soc. (EMBC)*, Jul. 2018, pp. 2535–2538.
- [84] Y. Luo, L.-Z. Zhu, Z.-Y. Wan, and B.-L. Lu, "Data augmentation for enhancing EEG-based emotion recognition with deep generative models," *J. Neural Eng.*, vol. 17, no. 5, Oct. 2020, Art. no. 056021.
- [85] L. F. Kozachenko and N. N. Leonenko, "Sample estimate of the entropy of a random vector," *Problemy Peredachi Informatsii*, vol. 23, no. 2, pp. 9–16, 1987.
- [86] A. Kraskov, H. Stögbauer, and P. Grassberger, "Estimating mutual information," *Phys. Rev. E, Stat. Phys. Plasmas Fluids Relat. Interdiscip. Top.*, vol. 69, no. 6, Jun. 2004, Art. no. 066138.
- [87] B. C. Ross, "Mutual information between discrete and continuous data sets," *PLoS ONE*, vol. 9, no. 2, Feb. 2014, Art. no. e87357.
- [88] E. Harmon-Jones, P. A. Gable, and C. K. Peterson, "The role of asymmetric frontal cortical activity in emotion-related phenomena: A review and update," *Biol. Psychol.*, vol. 84, no. 3, pp. 451–462, Mar. 2010.
- [89] J. A. Coan and J. J. B. Allen, "Frontal EEG asymmetry as a moderator and mediator of emotion," *Biol. Psychol.*, vol. 67, nos. 1–2, pp. 7–50, Oct. 2004.
- [90] H. J. Rosen, K. Pace-Savitsky, R. J. Perry, J. H. Kramer, B. L. Miller, and R. W. Levenson, "Recognition of emotion in the frontal and temporal variants of frontotemporal dementia," *Dementia Geriatric Cognit. Disorders*, vol. 17, no. 4, pp. 277–281, 2004.
- [91] H. J. Rosen, "Emotion comprehension in the temporal variant of frontotemporal dementia," *Brain*, vol. 125, no. 10, pp. 2286–2295, Oct. 2002.